

International Cybersecurity and Digital Forensics Academy (ICDFA)**SBT-DF203: Basic Networking Skills for Digital Forensics****HUSSEINI SULEIMAN****2025/FWSD/11478**fwsd2511478@students.icdfa.edu.ng**DATE: 22TH September, 2026****LAB 7: DNS Introduction and Traffic Analysis****Three-Week Delivery Block 3/3 | 19-25 September 2026**

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Important Pre-Lab Instruction

Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.

Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 7
Lab Title	DNS Introduction and Traffic Analysis
Delivery Block	3/3 of 3
Scheduled Dates	19-25 September 2026
Estimated Duration	3-4 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	dig_dns.pcap or locally generated DNS captures; optional smtp.pcap correlation

1. Source Materials and Slide Alignment

- Domain Name System Forensics: domain-to-IP resolution, hierarchical DNS, resolver/TLD/authoritative roles, dig queries, capture with tshark, query/response fields and DNS within email traffic.
- The slide practical captures dig and browser DNS traffic, examines A records, local resolver configuration, TTL and authoritative information.

- The exercise builds the normal DNS baseline required before analyzing spoofing in Lab 8.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

You are documenting normal DNS behaviour for an incident-response baseline. You will identify the configured resolver, capture controlled dig queries, analyze transaction IDs, names, record types, response codes, answers and TTL values, and correlate DNS resolution with a subsequent web or email connection.

3. Learning Outcomes

- Explain recursive and iterative DNS resolution at a foundational level.
- Use dig to query A, AAAA, MX and NS records.
- Capture and analyze DNS queries and responses with tshark/Wireshark.
- Identify transaction ID, query name, record type, response code, answer and TTL.
- Correlate DNS responses with later IP connections.
- Recognize normal variations such as multiple answers, CNAMEs, cache responses and IPv6 queries.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux or Ubuntu VM	Generate and analyze DNS traffic.
DNS utility	dig from dnsutils	Perform controlled queries.
Capture tools	tshark and Wireshark	Capture DNS traffic.
Evidence	dig_dns.pcap or new PCAPNG	Analysis source.
Resolver config	/etc/resolv.conf, resolvectl	Identify configured DNS service.
Optional browser	Firefox/Chrome	Generate multiple DNS lookups.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab7/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab7
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

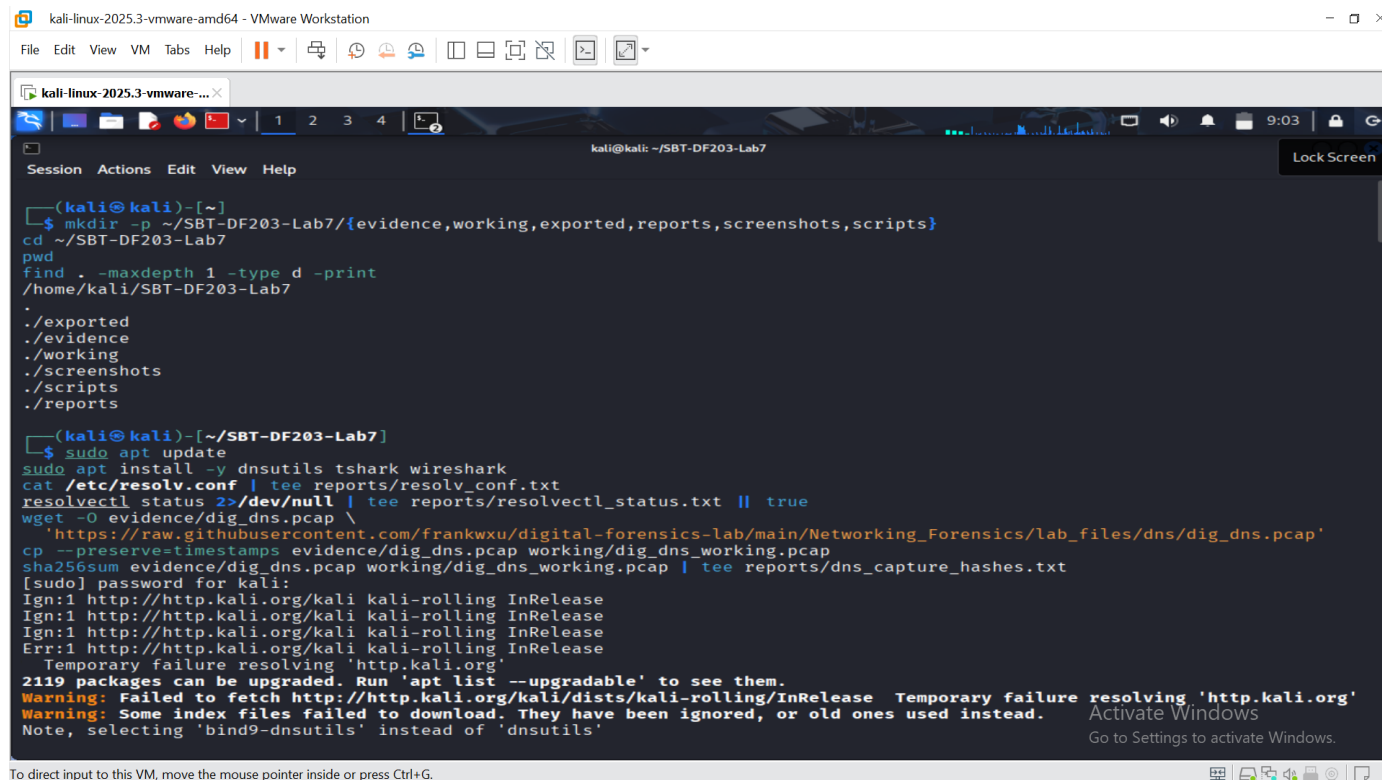
```
sudo apt update
sudo apt install -y dnsutils tshark wireshark
cat /etc/resolv.conf | tee reports/resolv_conf.txt
resolvectl status 2>/dev/null | tee reports/resolvectl_status.txt || true
wget -O evidence/dig_dns.pcap \
  'https://raw.githubusercontent.com/frankwxu/digital-forensics-
  lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap'
cp --preserve=timestamps evidence/dig_dns.pcap working/dig_dns_working.pcap
sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap | tee
reports/dns_capture_hashes.txt
```

Screenshot required: Configured resolver information, evidence details and hashes.

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab7-HUSSEINI SULEIMAN
Trainee name	HUSSEINI SULEIMAN
Date and time started	22/09/2026
Evidence file name(s)	dig_dns.pcap
Source or generation method	VM MACHINE
Original SHA-256	e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855 evidence/dig_dns.pcap
Working-copy SHA-256	e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855 working/dig_dns_working.pcap
Analysis workstation/VM	KALI@KALI AND ROOT@KALI
Notes on any changes	I CHANGE FROM KALI@KALI TO ROOT@KALI FOR SOME COMMAND NOT YIELDING RESULT ON KALI@KALI

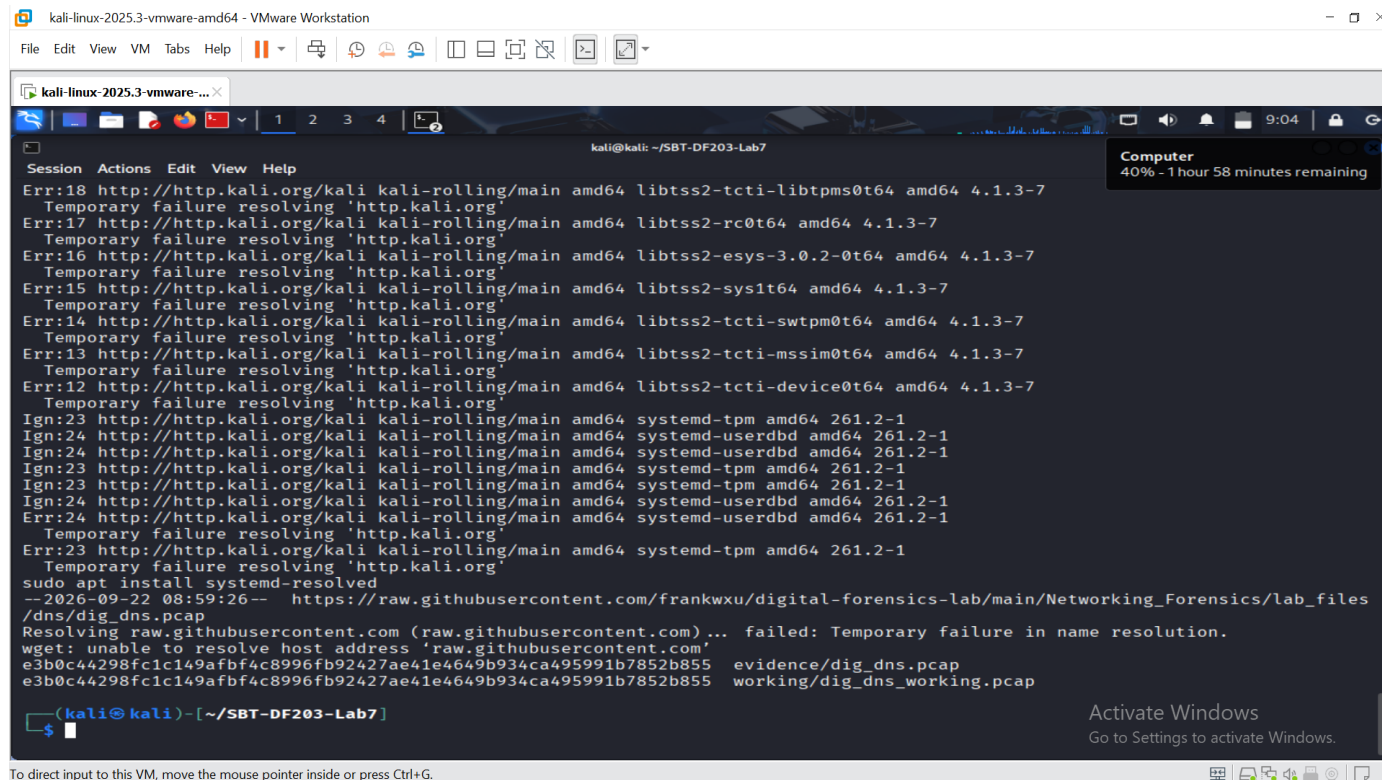
ANSWER: See the screenshot below



The screenshot shows a terminal window titled 'kali-linux-2025.3-vmware-amd64 - VMware Workstation'. The user is in the directory ~/SBT-DF203-Lab7. They run 'mkdir -p ~/SBT-DF203-Lab7/{evidence,working,exported,reports,screenshots,scripts}' and 'find . -maxdepth 1 -type d -print' to list the directories. Then they run 'sudo apt update' and 'sudo apt install -y dnsutils tshark wireshark'. The output shows the installation of dnsutils, tshark, and wireshark. The terminal also shows the contents of /etc/resolv.conf and the output of 'resolvectl status'. The terminal window has a 'Lock Screen' button in the top right corner.

```
(kali@kali)-[~]
$ mkdir -p ~/SBT-DF203-Lab7/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab7
pwd
/home/kali/SBT-DF203-Lab7
find . -maxdepth 1 -type d -print
.
./exported
./evidence
./working
./screenshots
./scripts
./reports

(kali@kali)-[~/SBT-DF203-Lab7]
$ sudo apt update
sudo apt install -y dnsutils tshark wireshark
cat /etc/resolv.conf | tee reports/resolv_conf.txt
resolvectl status 2>/dev/null | tee reports/resolvectl_status.txt || true
wget -O evidence/dig_dns.pcap \
'https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap'
cp --preserve-timestamps evidence/dig_dns.pcap working/dig_dns_working.pcap
sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap | tee reports/dns_capture_hashes.txt
[sudo] password for kali:
Ign:1 http://http.kali.org/kali kali-rolling InRelease
Ign:1 http://http.kali.org/kali kali-rolling InRelease
Ign:1 http://http.kali.org/kali kali-rolling InRelease
Err:1 http://http.kali.org/kali kali-rolling InRelease
Temporary failure resolving 'http.kali.org'
2119 packages can be upgraded. Run 'apt list --upgradable' to see them.
Warning: Failed to fetch http://http.kali.org/kali/dists/kali-rolling/InRelease Temporary failure resolving 'http.kali.org'
Warning: Some index files failed to download. They have been ignored, or old ones used instead.
Note, selecting 'bind9-dnsutils' instead of 'dnsutils'
```



The screenshot shows a terminal window titled 'kali-linux-2025.3-vmware-amd64 - VMware Workstation'. The user is in the directory ~/SBT-DF203-Lab7. They run 'sudo apt install systemd-resolved' and 'sudo apt install systemd-resolved'. The output shows the installation of systemd-resolved. The terminal also shows the output of 'resolvectl status' and the output of 'wget' and 'sha256sum'. The terminal window has a 'Computer' status bar in the top right corner showing '40% - 1 hour 58 minutes remaining'.

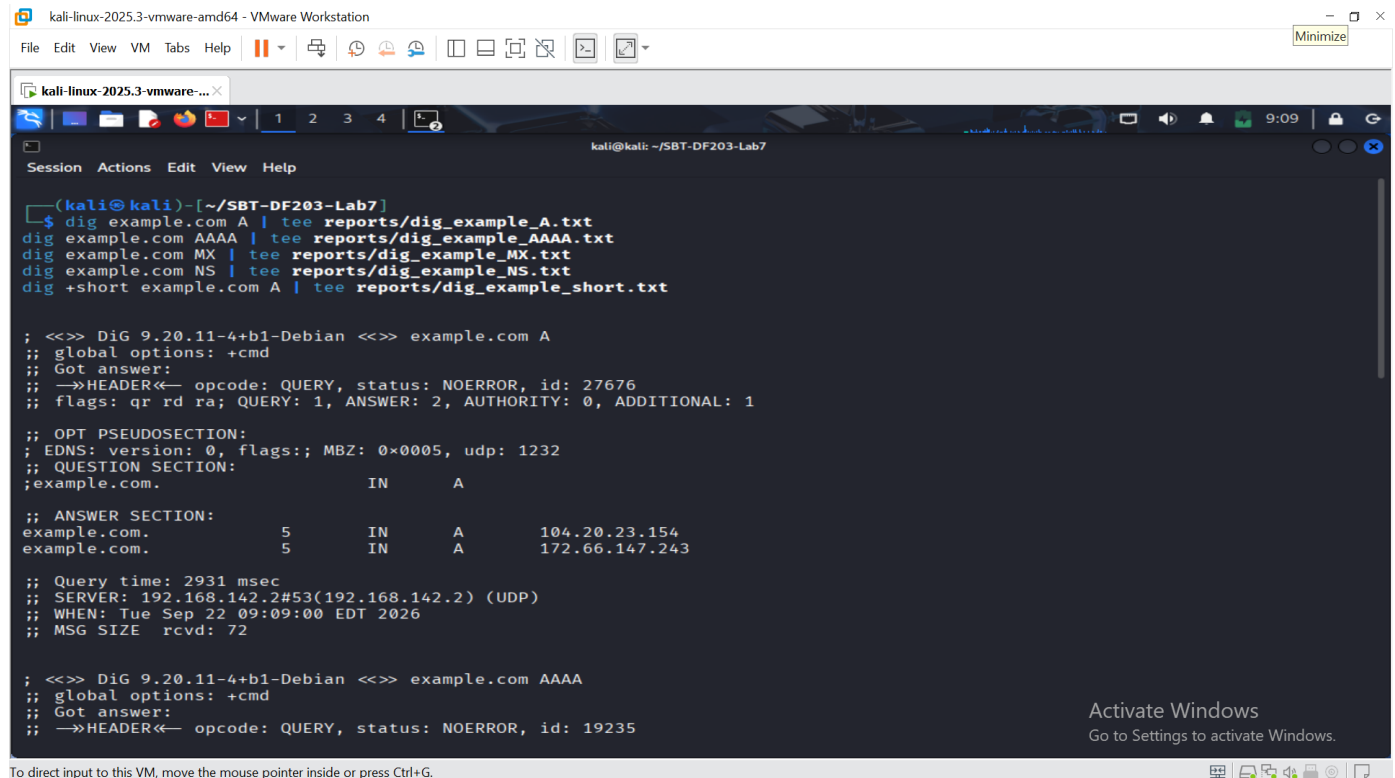
```
(kali@kali)-[~/SBT-DF203-Lab7]
$ sudo apt install systemd-resolved
--2026-09-22 08:59:26-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files
/dns/dig_dns.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com) ... failed: Temporary failure in name resolution.
wget: unable to resolve host address 'raw.githubusercontent.com'
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855 evidence/dig_dns.pcap
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855 working/dig_dns_working.pcap

(kali@kali)-[~/SBT-DF203-Lab7]
$
```

8. Part A - Query DNS Records with dig

```
dig example.com A | tee reports/dig_example_A.txt
dig example.com AAAA | tee reports/dig_example_AAAA.txt
dig example.com MX | tee reports/dig_example_MX.txt
dig example.com NS | tee reports/dig_example_NS.txt
dig +short example.com A | tee reports/dig_example_short.txt
```

ANSWER: SEE SCREENSHOT RESULT OF THE ABOVE PAYLOAD COMMAND



The screenshot shows a terminal window titled 'kali@kali: ~/SBT-DF203-Lab7'. The user has entered the following commands:

```
(kali@kali)-[~/SBT-DF203-Lab7]
$ dig example.com A | tee reports/dig_example_A.txt
dig example.com AAAA | tee reports/dig_example_AAAA.txt
dig example.com MX | tee reports/dig_example_MX.txt
dig example.com NS | tee reports/dig_example_NS.txt
dig +short example.com A | tee reports/dig_example_short.txt
```

The output of the first command is shown below:

```
; <<>> DiG 9.20.11-4+b1-Debian <<>> example.com A
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 27676
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:;, MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      A
;; ANSWER SECTION:
example.com.                  5       IN      A      104.20.23.154
example.com.                  5       IN      A      172.66.147.243

;; Query time: 2931 msec
;; SERVER: 192.168.142.2#53(192.168.142.2) (UDP)
;; WHEN: Tue Sep 22 09:09:00 EDT 2026
;; MSG SIZE rcvd: 72

; <<>> DiG 9.20.11-4+b1-Debian <<>> example.com AAAA
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 19235
```

An 'Activate Windows' watermark is visible in the bottom right corner of the terminal window.

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...X
kali@kali: ~/SBT-DF203-Lab7
Session Actions Edit View Help
;; ANSWER SECTION:
example.com.          5      IN      MX      0 .

;; Query time: 4595 msec
;; SERVER: 192.168.142.2#53(192.168.142.2) (UDP)
;; WHEN: Tue Sep 22 09:09:14 EDT 2026
;; MSG SIZE rcvd: 55

; <<>> DiG 9.20.11-4+b1-Debian <<>> example.com NS
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 56298
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      NS

;; ANSWER SECTION:
example.com.          5      IN      NS      elliot.ns.cloudflare.com.
example.com.          5      IN      NS      hera.ns.cloudflare.com.

;; Query time: 4079 msec
;; SERVER: 192.168.142.2#53(192.168.142.2) (UDP)
;; WHEN: Tue Sep 22 09:09:18 EDT 2026
;; MSG SIZE rcvd: 95

104.20.23.154
172.66.147.243

(kali@kali)~[~/SBT-DF203-Lab7]
$

```

Computer
41% - 1 hour 13 minutes until full

Activate Windows
Go to Settings to activate Windows.

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

- Record the resolver shown in the SERVER line: SERVER: 192.168.142.2#53(192.168.142.2) (UDP)
-
- Record status/response code, flags, question count, answer count and query time: WHEN: Tue Sep 22 09:09:18 EDT 2026
- Explain IN, A, AAAA, MX and NS.
- Record at least one TTL and explain that it is a cache lifetime, not a creation date: EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232

ANSWER:

9. Part B - Capture a Fresh DNS Query

IFACE=eth0

sudo tshark -i "\$IFACE" -f 'port 53' -a duration:25 -w evidence/fresh_dig_dns.pcapng & sleep 3

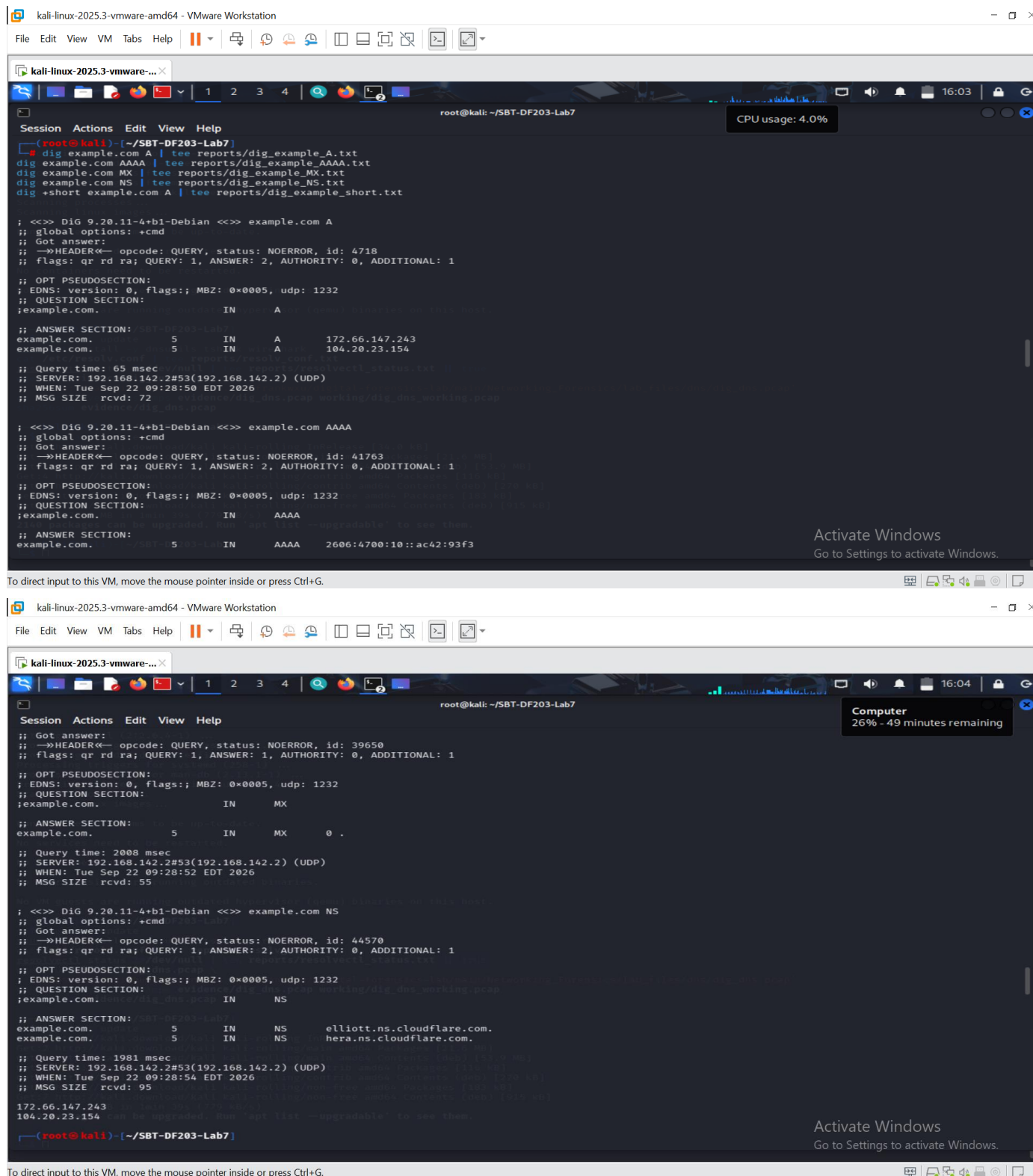
dig +noedns example.com A >/dev/null

wait

sha256sum evidence/fresh_dig_dns.pcapng | tee reports/fresh_dns_sha256.txt

Interface Note

On systems using a local stub resolver, packets may appear on lo and then on a physical interface. Use `tshark -D` to list interfaces and choose the one that shows the actual query.



The screenshot shows a Kali Linux terminal window with the following content:

```
root@kali: ~/SBT-DF203-Lab7
CPU usage: 4.0%

Session Actions Edit View Help
root@kali) [~/SBT-DF203-Lab7]
# dig example.com A | tee reports/dig_example_A.txt
dig example.com AAAA | tee reports/dig_example_AAAA.txt
dig example.com MX | tee reports/dig_example_MX.txt
dig example.com NS | tee reports/dig_example_NS.txt
dig +short example.com A | tee reports/dig_example_short.txt

; <<>> DiG 9.20.11-4+b1-Debian <<>> example.com A
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 4718
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      A
;; ANSWER SECTION:
example.com. 5 IN A 172.66.147.243
example.com. 5 IN A 104.20.23.154
;; Query time: 65 msec
;; SERVER: 192.168.142.2#53(192.168.142.2) (UDP)
;; WHEN: Tue Sep 22 09:28:50 EDT 2026
;; MSG SIZE rcvd: 72

; <<>> DiG 9.20.11-4+b1-Debian <<>> example.com AAAA
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 41763
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      AAAA
;; ANSWER SECTION:
example.com. 5 IN AAAA 2606:4700:10::ac42:93f3
;; Query time: 65 msec
;; SERVER: 192.168.142.2#53(192.168.142.2) (UDP)
;; WHEN: Tue Sep 22 09:28:50 EDT 2026
;; MSG SIZE rcvd: 72

; <<>> DiG 9.20.11-4+b1-Debian <<>> example.com MX
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 39650
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      MX
;; ANSWER SECTION:
example.com. 5 IN MX 0 .
;; Query time: 2008 msec
;; SERVER: 192.168.142.2#53(192.168.142.2) (UDP)
;; WHEN: Tue Sep 22 09:28:52 EDT 2026
;; MSG SIZE rcvd: 55

; <<>> DiG 9.20.11-4+b1-Debian <<>> example.com NS
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 44570
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      NS
;; ANSWER SECTION:
example.com. 5 IN NS elliot.ns.cloudflare.com.
example.com. 5 IN NS hera.ns.cloudflare.com.
;; Query time: 1981 msec
;; SERVER: 192.168.142.2#53(192.168.142.2) (UDP)
;; WHEN: Tue Sep 22 09:28:54 EDT 2026
;; MSG SIZE rcvd: 95

172.66.147.243
104.20.23.154

root@kali) [~/SBT-DF203-Lab7]
```

Activate Windows
Go to Settings to activate Windows.

Computer
26% - 49 minutes remaining

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...
Session Actions Edit View Help
Desktop - Thunar @kali: ~/SBT-DF203-Lab7
(kali@kali)-[~/SBT-DF203-Lab7]
$ IFACE=eth0
sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w evidence/fresh_dig_dns.pcapng &
sleep 3
dig +noedns example.com A >/dev/null
wait
sha256sum evidence/fresh_dig_dns.pcapng | tee reports/fresh_dns_sha256.txt
[6] 123282
[sudo] password for kali:
[6] + suspended (tty input) sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w
60a3d590cb67a5ab481133efc48213240f88ba28f566a5e759afcd58b2942ab3 evidence/fresh_dig_dns.pcapng
(kali@kali)-[~/SBT-DF203-Lab7]
$

```

Activate Windows
Go to Settings to activate Windows.

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

10. Part C - Extract DNS Query and Response Fields

PCAP=evidence/fresh_dig_dns.pcapng

Query

```
tshark -r "$PCAP" -Y 'dns.flags.response==0' -T fields \
-e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
-e dns.id -e dns.qry.name -e dns.qry.type \
| tee reports/dns_queries.tsv
```

Response

```
tshark -r "$PCAP" -Y 'dns.flags.response==1' -T fields \
-e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
-e dns.id -e dns.flags.rcode -e dns.count.answers -e dns.a -e dns.aaaa -e dns.resp.ttl \
| tee reports/dns_responses.tsv
```

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help

kali-linux-2025.3-vmware-...
root@kali: ~/SBT-DF203-Lab7

Session Actions Edit View Help
104.20.23.154

(root@kali)~[~/SBT-DF203-Lab7]
# IFACE=eth0
sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w evidence/fresh_dig_dns.pcapng &
sleep 3
dig +noedns example.com A >/dev/null
wait
sha256sum evidence/fresh_dig_dns.pcapng | tee reports/fresh_dns_sha256.txt

[1] 51016
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'
2
[1] + done sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w
90082306b30eeaff96ff76785b87e6fef51ba8551ea4ffae1516e80dd545aa28 evidence/fresh_dig_dns.pcapng

(root@kali)~[~/SBT-DF203-Lab7]
# PCAP=evidence/fresh_dig_dns.pcapng
# Query
tshark -r "$PCAP" -Y 'dns.flags.response==0' -T fields \
-e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
-e dns.id -e dns.qry.name -e dns.qry.type \
| tee reports/dns_queries.tsv

# Response
tshark -r "$PCAP" -Y 'dns.flags.response==1' -T fields \
-e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
-e dns.id -e dns.flags.rcode -e dns.count.answers -e dns.a -e dns.aaaa -e dns.resp.ttl \
| tee reports/dns_responses.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
1 2026-09-22T09:29:43.697328540-0400 192.168.142.128 48932 192.168.142.2 53 0x3036 example.com 1
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
2 2026-09-22T09:29:43.703364035-0400 192.168.142.2 53 192.168.142.128 48932 0x3036 0 2 104.20.23.154,172.66.147.243 5

(root@kali)~[~/SBT-DF203-Lab7]
Activate Windows
Go to Settings to activate Windows.

```

11. Part D - Match Queries to Responses

Transaction ID	Query Name	Type	Client/Resolver	Response Code	Answer(s)	TTL	Time Delta
4718	example.com A	A	172.66.147.243	NOERROR	172.66.147.243 104.20.23.154	5	65 msec
41763	example.com AAAA	AAAA	192.168.142.2	NOERROR,	2606:4700:10::a c42:93f3	5	53 msec
39650	example.com MX	MX	192.168.142.2	NOERROR,	192.168.142.2#53(192.168.142.2)	5	2008 msec

- Match query and response using transaction ID plus the endpoint tuple.

- Confirm that the response source is the queried resolver.
- Calculate response time from frame timestamps.
- Document CNAME chains and multiple A/AAAA answers where present.

12. Part E - Analyze DNS Generated by a Browser

A browser may request several domains for the page, images, fonts, telemetry or cached services. Capture a visit to a non-sensitive instructor-approved site and inventory all query names.

```
sudo tshark -i IFACE -f 'port 53' -a duration:40 -w evidence/browser_dns.pcapng &  
sleep 3
```

```
# Open the approved site in a private browser window, then wait.  
wait
```

```
tshark -r evidence/browser_dns.pcapng -Y 'dns.flags.response==0' -T fields -e  
dns.qry.name -e dns.qry.type \  
| sort | uniq -c | sort -nr | tee reports/browser_dns_inventory.txt
```

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help

kali-linux-2025.3-vmware-...
root@kali: ~/SBT-DF203-Lab7
CPU usage: 0.0%

# Query
tshark -r "$PCAP" -Y 'dns.flags.response==0' -T fields \
-e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
-e dns.id -e dns.qry.name -e dns.qry.type \
| tee reports/dns_queries.tsv

# Response
tshark -r "$PCAP" -Y 'dns.flags.response==1' -T fields \
-e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
-e dns.id -e dns.flags.rcode -e dns.count.answers -e dns.a -e dns.aaaa -e dns.resp.ttl \
| tee reports/dns_responses.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
1 2026-09-22T09:29:43.697328540-0400 192.168.142.128 48932 192.168.142.2 53 0*3036 example.com 1
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
2 2026-09-22T09:29:43.703364035-0400 192.168.142.2 53 192.168.142.128 48932 0*3036 0 2 104.20.23.154,172.66.147.243 5
.5

(root@kali) - [~/SBT-DF203-Lab7]
# sudo tshark -i IFACE -f 'port 53' -a duration:40 -w evidence/browser_dns.pcapng &
sleep 3
# Open the approved site in a private browser window, then wait.
wait

tshark -r evidence/browser_dns.pcapng -Y 'dns.flags.response==0' -T fields -e dns.qry.name -e dns.qry.type \
| sort | uniq -c | sort -nr | tee reports/browser_dns_inventory.txt

[1] 58279
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on "IFACE"
tshark: There is no device named "IFACE".
(No such device exists)
[1] + exit 1 sudo tshark -i IFACE -f 'port 53' -a duration:40 -w
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
tshark: The file "evidence/browser_dns.pcapng" doesn't exist.

(root@kali) - [~/SBT-DF203-Lab7]
Activate Windows
Go to Settings to activate Windows.

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

13. Part F - Correlate DNS with Subsequent Connections

Extract DNS A answers

```
tshark -r evidence/browser_dns.pcapng -Y 'dns.a' -T fields -e frame.time_epoch -e
dns.qry.name -e dns.a \
| tee reports/dns_A_answers.tsv
```

Extract later TCP destinations

```
tshark -r evidence/browser_dns.pcapng -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T
fields \
-e frame.time_epoch -e ip.dst -e tcp.dstport \
| tee reports/subsequent_tcp_destinations.tsv
```

Select one DNS answer and show whether the client later initiates a connection to that IP. Explain any mismatch caused by multiple answers, proxies, CDNs, IPv6, caching or capture duration.

14. Part G - DNS Analysis Within SMTP Evidence

If smtp.pcap from Lab 4 is available, filter dns and identify the query used to locate a mail server. Record the queried name, record type, resolver, answer and TTL.

```
# Optional, adjust path to your Lab 4 capture
tshark -r ../SBT-DF203-Lab4/working/smtp_working.pcap -Y 'dns' -T fields \
  -e frame.number -e frame.time -e dns.qry.name -e dns.qry.type -e dns.a -e dns.resp.name \
  -e dns.resp.ttl \
  | tee reports/smtp_dns_correlation.tsv
```

```
kali@kali: ~/SBT-DF203-Lab7
$ # Optional, adjust path to your Lab 4 capture
tshark -r ../SBT-DF203-Lab4/working/smtp_working.pcap -Y 'dns' -T fields \
  -e frame.number -e frame.time -e dns.qry.name -e dns.qry.type -e dns.a -e dns.resp.name -e dns.resp.ttl \
  | tee reports/smtp_dns_correlation.tsv
Warning: program compiled against libxml 215 using older 214
1      2009-10-05T02:06:07.492060000-0400      mail.patriots.in      1      74.53.140.153      mail.patriots.in,patriots.in,patriots.in,patriots.in      1082
2      2009-10-05T02:06:07.526085000-0400      mail.patriots.in      1
7,10828,82828,82828
(kali@kali)~[~/SBT-DF203-Lab7]
$
```

15. Required Findings Worksheet

Field	Finding
Configured resolver IP	74.53.140.1
Client source port:	53
Resolver destination port:	43
Transaction ID	:44570
Query name and type:	elliott.ns.cloudflare.com.
Response code:	NOERROR
Answer IP(s): SERVER:	192.168.142.2#53(192.168.142.2) (UDP)
TTL: 5	5

Field	Finding
Query-response time delta	: 1981 msec
Subsequent connection correlation	The same query has different ip address. Eg: 172.66.147.243 104.20.23.154
Normal variations observed	Query type differs from A,AAAA,NS and mx.

Required Screenshots Checklist

- ☐ Resolver configuration.
- ☐ dig A/AAAA/MX/NS outputs.
- ☐ Fresh DNS capture hash.
- ☐ DNS query fields.
- ☐ Matching DNS response fields.
- ☐ Transaction ID correlation.
- ☐ Answer and TTL.
- ☐ Browser query inventory.
- ☐ DNS answer to TCP destination correlation.
- ☐ Optional SMTP DNS evidence.

Student Report Template

1. Executive summary.
2. Resolver and environment.
3. Evidence integrity.
4. dig results.
5. Packet-level query/response analysis.
6. Transaction matching and timing.
7. Browser DNS inventory.
8. Connection correlation.
9. Limitations and normal variations.
10. Appendix.

Submission Requirements

- One PDF report named SBT-DF203-Lab7_RegNo_FullName.pdf.
- The original or generated PCAP/PCAPNG evidence file and its SHA-256 hash text file.

- A reports folder containing exported CSV/TXT command output and any recovered objects.
- Screenshots must be readable, numbered and referenced in the report narrative.
- Compress the report and evidence outputs into one ZIP named with your registration number and lab number.
- Do not submit passwords or decoded credentials in public discussion channels; include them only in the protected assessment submission where required.

Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Preparation and resolver identification	10	Correct resolver and evidence hashes.
dig record analysis	15	A, AAAA, MX, NS and TTL interpretation.
Packet-field extraction	25	Correct IDs, names, types, endpoints and response codes.
Query-response correlation	20	Accurate matching and timing.
Browser/connection correlation	20	Inventory and defensible IP connection linkage.
Report quality	10	Clear screenshots, tables and conclusions.

Instructor Quick Answer Guide

Checkpoint	Expected Observation
Standard transport	Usually UDP destination port 53; TCP may be used for large responses or zone transfer.
Query flag	<code>dns.flags.response==0</code> .
Response flag	<code>dns.flags.response==1</code> .
A record	IPv4 address.
AAAA record	IPv6 address.

Checkpoint	Expected Observation
TTL	How long a resolver may cache the record.

Command Reference Summary

Command	Purpose
dig NAME TYPE	Query a DNS record.
dns.qry.name	Queried domain name.
dns.qry.type	Requested record type.
dns.id	DNS transaction identifier.
dns.flags.rcode	Response code.
dns.resp.ttl	Answer TTL.

Academic Integrity and Authorization Statement

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.